Shadow API Risk Detection using Machine Learning

Introduction

APIs are essential for communication between software systems, but undocumented or unmanaged APIs, known as shadow APIs, can create serious security risks. These APIs may expose sensitive information and increase vulnerabilities because they are not properly monitored. This project uses machine learning techniques to detect risky API calls automatically and support API security monitoring.

Objective

The main objective of this project is to build a machine learning model that can identify and classify shadow API calls based on their behavior. The project also compares different classification algorithms to evaluate their performance.

Dataset and Tools Used

A supervised API behavior dataset was used for training and testing the models. The implementation was done using Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn and Jupyter Notebook.

Methodology

The dataset was first cleaned by removing unnecessary columns and checking missing values. Categorical features were converted into numerical values, and the data was split into training and testing sets. Two machine learning models were applied:

- Logistic Regression
- Random Forest

The models were evaluated using accuracy, classification report, confusion matrix.

Results

Logistic Regression achieved **99.7% accuracy**, with only one normal API call misclassified. Random Forest achieved very strong classification performance and correctly identified almost all API calls. Both models showed reliable training and testing performance.

Conclusion

This project shows that machine learning can be effectively used to detect risky shadow API calls. The developed solution can help organizations improve API security by automatically identifying suspicious API behavior.